



Credibility & Newsworthiness

Web Science
COMPSCI4077/COMPSCI5107/COMPSCI5078

Joemon M Jose

Overall info

announcements

Communication

- Through Teams Page
 - so please sign in -
 - [COMPSCI4077, COMPSCI5078 & COMPSCI5107: WEB SCIENCE - 2023-24 | General | Microsoft Teams](#)
 - If you are not on Teams Channel
 - you will not get any information/emails
- Lecture materials
 - Moodle
 - <https://moodle.gla.ac.uk/course/view.php?id=44995>
- Coursework & deadlines
 - Announced in Teams File Area – Files/class materials/courseworks/
 - Read them – geolocalisation lecture third week – Reddit Lecture on week 4 – ask questions during the lecture
 - Deadline (TBC by LTC)
 - Level H – GeoLocalisation - 28th February
 - Level M – Reddit Data Analysis - 7th March

Which module to enroll?

- COMPSCI4077 - Web Science H – For level H students
- COMPSCI5107 - Web Science for MSc
- COMPSCI5078 - Web Science M – For Level M students
- Please enrol for the correct module
 - That fit with your course plan

Please watch the Introduction lecture

- If you have any questions, please write your questions at
 - https://glamys.sharepoint.com/personal/joemon_jose_glasgow_ac_uk/_layouts/15/doc.aspx?sourcedoc={9185079d-37bf-48dd-84db-aa34e6aeb910}&action=edit
 - I will answer them in the class
 - This will help rest of the class as well
 - they will get an opportunity to clarify concepts

Exams

- Follow the rules of School of Computing Science
 - Which is based on the University of Glasgow Guidelines
- Online and Supervised
 - Openbook
- Hence, solving similar but open problems
 - Need to have a better understanding of the topics so that
 - You should be able to apply solutions intelligently

Last lecture

- Social Media data
 - Large amount of data is generated
 - Potential to use as a social sensor
- Two types of tweets
 - bragging, marketing – noisy tweets
 - informative
- Tweet text processing

Learning Objectives

- What is newsworthiness?
- How to identify them automatically?
- Distant supervision approach

Elites who work behind the scenes

Who contribute time and money

To every single Elite who makes Erica feel special and loved
God bless you 🙏🙏🙏❤️ #AkwaabaEricaNlewedim



@gavincost @Freedom16356531 I mean he's not wrong about the alarming hairline 😂
@gavincost @Freedom16356531 I mean **he s** not wrong about the alarming hairline



Let us look at some tweet texts

- GaryMan65617019
 - @Sky
- halomaryam
 - im sorry its like 7am in london rn smhNews Stop bullying UK!
- GoldJosh4
 - @SkyNews Mind blowing offer just click here
 - <https://t.co/BdfxGXxkWk>
- kloker_heather
 - mobil porn uk milf
 - @sex_porn_linkk
- SplashThatCash1
 - LUXURY CRUSHED VELVET EYELET CURTAINS
 - Transform and modernise your room with the crushed velvet curtains.
 - Direct Ordering & Direct UK Delivery 🚚
 - 7-14 working day delivery 🚚💖
- SkyNews
 - Brexit trade deal deadline missed as Sturgeon calls for transition period extension <https://t.co/pKmjOgaVDZ>
- SubwayUK
 - @hannahox We are sorry to hear about this situation. Due to privacy regulations, we are not able to look into this. Please reach out to our colleagues from Subway Rewards at <https://t.co/zYvH9HFfYR>. The team will be able to assist you. Thank you.
- NidhiKNDTV
 - We need to be even more cautious now....halt flights for a few days!
 - #Covid_19
 - #CoronavirusMutation
- guardian
 - Sokratis free to find new club after Arsenal cancel defender's contract <https://t.co/atx56ibXM1>
- KLM_UK
 - @paajuuzaidi You can find all the contact details of London City airport here: <https://t.co/CiGRleDWs>.

Spam

Marketing

No usefulness!

Can we remove noisy text?

- Spammers
 - @SkyNews Mind blowing offer just click here
 - <https://t.co/BdfxGXxkWk>
- Marketers
 - LUXURY CRUSHED VELVET EYELET CURTAINS
 - Transform and modernise your room with the crushed velvet curtains.
 - Direct Ordering & Direct UK Delivery 🚚
 - 7-14 working day delivery 🚚💖
- No substance
 - im sorry its like 7am in london rn smhNews Stop bullying UK!
- But
 - How?
- Removing noisy texts
 - Helps in downstream applications like event detection
 - Event detection approaches (future lecture) will work on
 - require fewer tweets

Modelling *interestingness* or newsworthiness of tweets!

- Challenges
 - Unstructured tweets
 - Often composed in short time, not much editorial effort
 - Informal nature
 - Friendly communications
 - Unpredictable nature of news
 - **Each newsworthy tweet is different**
 - WMerciaPolice - If you're offered unusually high returns on an investment, it could be a scam.
 - MetroRadioNews - Boris Johnson's been chairing a Cobra crisis meeting this afternoon as #StormChristoph continues to hit the UK. We have several flood alerts in the North East. Last night, a man had to be rescued after his delivery van overturned in County Durham:

Newsworthy? What is it?

- **A tweet is newsworthy**
 - **If it discusses a topic that is of interest to news media**
 - Newsworthiness should be measurable
- Example
 - lond0n4
 - @MrMonsta6 Where in London?
 - emma_angel3
 - New York attorney general sues NYPD over "illegal and harmful conduct" at racial justice protests <https://t.co/mc8OyYscQD>
 - PublicHealthW
 - @DavidTCDavies Public Health Wales has not denied access to the Covid-19 vaccine to any care home residents. The Welsh Government is leading on the deployment of the vaccine in Wales. In supporting them Public Health Wales has been planning for the vaccination of priority groups, [...]
 - SunManUtd
 - Premier League to show all fixtures live in UK until end of February as chiefs plan for rest of season without fans <https://t.co/qkWhuWYegg>
- Scale
 - Less relevant to highly relevant

Key requirements of a newsworthy score

- Real-time
 - Tweets should be scored as soon as it arrives!
- Generalizability
 - Should be able to handle any types of events – not those just seen before
 - Traffic accident Airport disruption ... power failure
- Adaptive
 - New information to be incorporated, as and when they arrive

['chiefs', 'end', 'fans', 'february', 'fixtures', 'https://t.co/qkWhuWYegg', 'league', 'live', 'plan', 'premier', 'rest', 'se
'show', 'without']

['#MUFC', '2021', 'confirmed', 'fans', 'february', 'league', 'live', 'matches', 'official', 'premier', 'shown']

['announced', 'end', 'february', 'games', 'https://t.co/p33KhgTLEQ', 'january', 'league', 'live', 'premier', 'shown']

- How to incorporate new information to the scoring model?

How do we do this?

- Classification approach
 - If we have a manually annotated data set, ..
- Distant supervision
 - Training data is identified in real-time using a set of heuristics

UnitedsUpdate

Manchester United are seeking buyers for Marcos Rojo and Sergio Romero this month and have valued both players at m
ethicsinthenews

The Oxford Uehiro Centre's Tess Johnson discusses genetic enhancement with @OxfordSparks - can it help us stick to our
ivadixit

I hope all the people who followed me thinking I'm a Serious New York Times blue check who gets nazi threats in the mai
PublicHealthW

@DavidTCDavies Public Health Wales has not denied access to the Covid-19 vaccine to any care home residents. The Wel
ianguides

@CliffTaylorIT One thing I've consistently said more months now, at individual level €350 isn't a lot to exist on especially
MirrorCeleb

Jennifer Garner breaks cover after her ex Ben Affleck and Ana de Armas split <https://t.co/m0Tol4BmCE>

Potentially low-quality text

theetryagains1

This is eating lowkey <https://t.co/1Vzqlg2NGu>

GaneshY69613050

@mohsinaliisb @razi_haider I have one question

England test with Rohit which pair is best suited for open, Gill, rahul, agrawal, because rahul having good record with

BillyFlood20

PERFECTION

bosun_higgs

@annmari38349484 @grahamlaurie1 @agcolehamilton 'The ability to put a financial package in place...'

Scotland would be broke by this point. It's taken a year to get a vaccine, also to be procured independently.

Locking down international travel stops the flow of goods...you've not really thought this through, have you?

Army_Tanker83

@Rachael56152878 @Celinagreen09 @JackPosobiec As a combat vet that has deployed and never been vetted I can

cashcartiisbae

in new york i really rock , ask playboi carti

Heuristic Labelling

- Semi-automatic labelling approach
 - Using a set of heuristics to label
 - High quality (newsworthy) and low quality (noisy) content
 - This will not label majority of the content
 - However, still create enough data to model quality of data
- Advantages
 - Minimal effort in creating a data set
 - Real-life data set – incremental and generalizable
 - Easily built as part of an algorithm for example event detection

Overall approach

- Collect a set of high-quality sets and
- low-quality sets of data
- Use this dataset to potentially score a newsworthy tweet

Title **Real-time event detection using Twitter / Andrew James McMinn.**
Author [McMinn, Andrew James, author.](#)
Production [Glasgow] : University of Glasgow, 2018.

Chapter 5

<https://eleanor.lib.gla.ac.uk/record=b3335434>

Features to use

Part - 2

```

"user": {
  "id": 278177631,
  "id_str": "278177631",
  "name": "David O'Regan 📖📰",
  "screen_name": "David_uhan",
  "location": "England, UK",
  "url": "http://www.linkedin.com/in/DirectorRCSEdFstMedicineGov",
  "description": "Past Director FST @RCSEd - cardiac surgeon Executive Director @MEDICINEGOV #BBASS #TeamPatient #MedLearn #TeamSurgical Council @HCSA",
  "translator_type": "none",
  "protected": false
}

```

descriptionWeight of the profile

listTerms = ['news', 'report', 'journal', 'write', 'editor', 'analyst', 'analysis', 'media', 'updates', 'stories', 'trader', 'investor', 'forex', 'stock', 'finance', 'market']

- News & Journalism
- Weight = 2.0
- Journalists, news organizations, financial traders use in their user description on Twitter
- To provide high coverage, we can use root forms
 - report -> report, reporters, reports etc.

listSpam = ['ebay', 'review', 'shopping', 'deal', 'sale', 'sales', 'link', 'click', 'marketing', 'promote', 'discount', 'products', 'store', 'diet', 'weight', 'porn', 'followback', 'follow back', 'lucky', 'winners', 'prize', 'hiring']

- Weight = 0.1
- Based on manual analysis

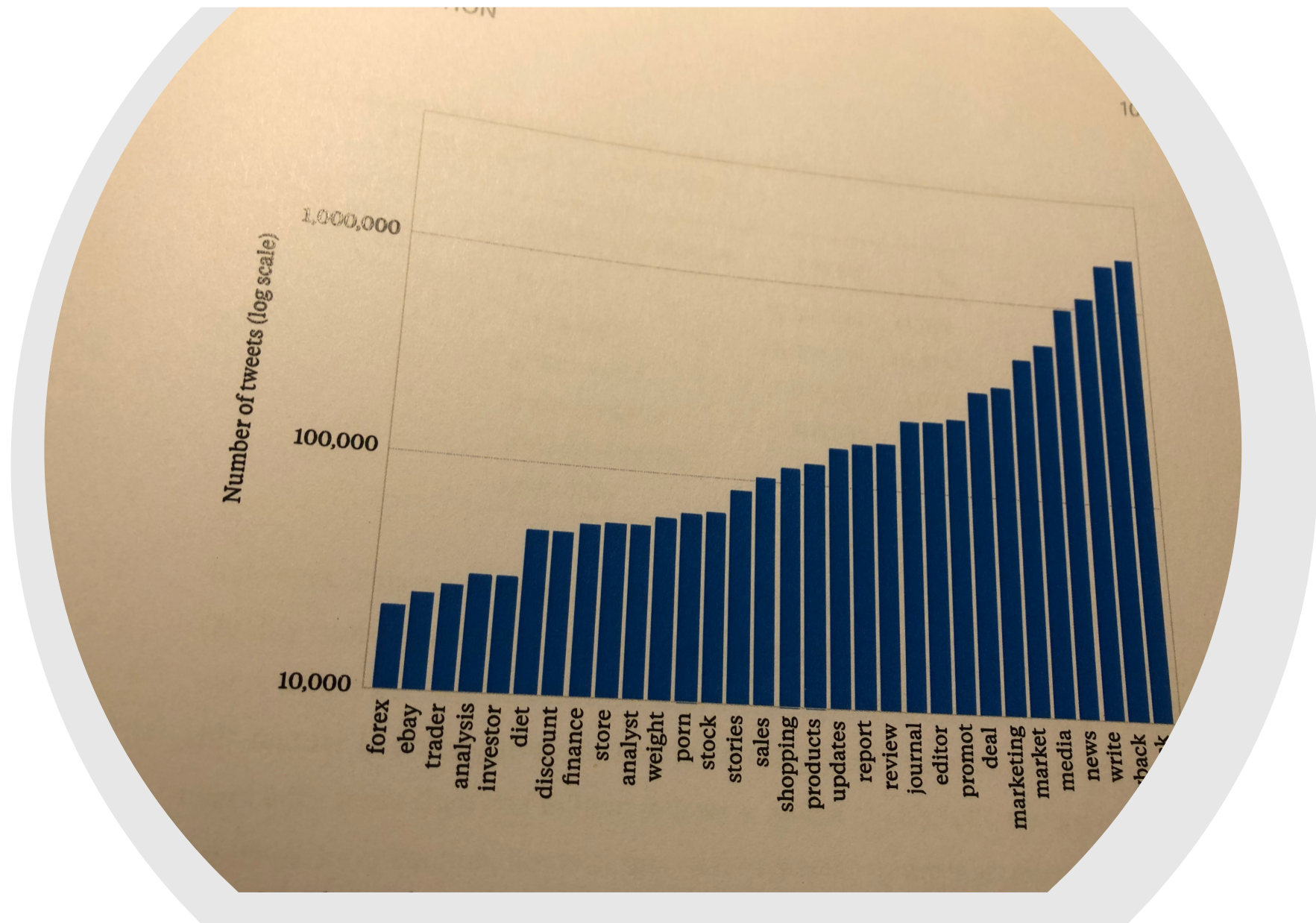
Normalise based on potential maxWeight

- Sum of termWeights / maxWeight
- Range between [0 and 1]

For all other terms
weight = 1.0

maxWeight

- What is the potential maxWeight?
- If all terms are in the **listTerms**, then
 - **maxWeight = 2 x length of the description**



Change in terms

- List of spam changes periodically
-
- **listSpam** = ['ebay', 'review', 'shopping', 'deal','sale', 'sales','link', 'click', 'marketing', 'promote', 'discount', 'products', 'store', 'diet', 'weight', 'porn', 'followback', 'follow back', 'lucky', 'winners', 'prize', 'hiring']

Account Age: accountAgeWeight

- Recent accounts are generally viewed as less credible than
 - accounts that have been active for longer periods of time
- daysSince = number of days since account is created

if daysSince < 1:

weight = 0.05

elif daysSince < 30:

weight = 0.10

elif daysSince < 90:

weight = 0.25

elif daysSince > 90:

weight = 1.0

accountAgeWeight = weight

(0..1]

```
13 "user": {
14   "id": 1161317586,
15   "id_str": "1161317586",
16   "name": "Nathan Fry",
17   "screen_name": "NathanFryLeeds",
18   "location": null,
19   "url": null,
20   "description": null,
21   "translator_type": "none",
22   "protected": false,
23   "verified": false,
24   "verified_type": "blue",
25   "followers_count": 119,
26   "friends_count": 299,
27   "listed_count": 1,
28   "favourites_count": 49023,
29   "statuses_count": 13594,
30   "created_at": "Fri Feb 08 22:00:57 +0000 2013",
31   "utc_offset": null,
32   "time_zone": null,
33   "geo_enabled": false,
34   "lang": null,
35   "contributors_enabled": false,
36   "is_translator": false,
37   "profile_background_color": "C0DEED",
38   "profile_background_image_url": "http://abs.twimg.com/images/themes/theme1/bg.png",
39   "profile_background_image_url_https": "https://abs.twimg.com/images/themes/theme1/bg.png",
40   "profile_background_tile": false,
41   "profile_link_color": "1DA1F2",
42   "profile_sidebar_border_color": "C0DEED",
43   "profile_sidebar_fill_color": "DDEEFF",
44   "profile_text_color": "333333",
45   "profile_use_background_image": true,
46   "profile_image_url": "http://pbs.twimg.com/profile_images/1544611925488705539/jUKIH2Q_normal.jpg",
47   "profile_image_url_https": "https://pbs.twimg.com/profile_images/1544611925488705539/jUKIH2Q_normal.jpg",
48   "profile_banner_url": "https://pbs.twimg.com/profile_banners/1161317586/1648019074",
49   "default_profile": true,
50   "default_profile_image": false,
51   "following": null,
52   "follow_request_sent": null,
53   "notifications": null,
54   "withheld_in_countries": []
55 }
```



Data distribution

5.3. EVALUATION

Table 5.6: The number of tweets in the collection (excluding retweets) for various volumes of tweets per day, on average.

Account Age	Tweets	%
< 1 day	786,989	1.05%
1-29 days	2,286,548	3.04%
30-90 days	3,863,935	5.14%
> 90 days	68,239,524	90.77%

Number of Followers : followersWeight

- Something becomes news because who said it rather than what is said
 - Head of state, public figure, celebrity
- Number of followers a user has can infer

if followersCount < 50:

weight = 0.5

elif followersCount < 5000:

weight = 1.0

elif followersCount < 10000:

weight = 1.5

elif followersCount < 100000:

weight = 2.0

elif followersCount < 200000:

weight = 2.5

elif followersCount > 200000:

weight = 3.0

followersWeight= weight/3

(0...1]



```
13  "user": {
14    "id": 1161317586,
15    "id_str": "1161317586",
16    "name": "Nathan Fry",
17    "screen_name": "NathanFryLeeds",
18    "location": null,
19    "url": null,
20    "description": null,
21    "translator_type": "none",
22    "protected": false,
23    "verified": false,
24    "verified_type": "blue",
25    "followers_count": 119,
26    "friends_count": 299,
27    "listed_count": 1,
28    "favourites_count": 49023,
29    "statuses_count": 13594,
30    "created_at": "Fri Feb 08 22:00:57 +0000 2013",
31    "utc_offset": null,
32    "time_zone": null,
33    "geo_enabled": false,
34    "lang": null,
35    "contributors_enabled": false,
36    "is_translator": false,
37    "profile_background_color": "C0DEED",
38    "profile_background_image_url": "http://abs.twimg.com/images/themes/theme1/bg.png",
39    "profile_background_image_url_https": "https://abs.twimg.com/images/themes/theme1/bg.png",
40    "profile_background_tile": false,
41    "profile_link_color": "1DA1F2",
42    "profile_sidebar_border_color": "C0DEED",
43    "profile_sidebar_fill_color": "DDEEF6",
44    "profile_text_color": "333333",
45    "profile_use_background_image": true,
46    "profile_image_url": "http://pbs.twimg.com/profile_images/1544611925488705539/jUKIHhz0_normal.jpg",
47    "profile_image_url_https": "https://pbs.twimg.com/profile_images/1544611925488705539/jUKIHhz0_normal.jpg",
48    "profile_banner_url": "https://pbs.twimg.com/profile_banners/1161317586/1648019074",
49    "default_profile": true,
50    "default_profile_image": false,
51    "following": null,
52    "follow_request_sent": null,
53    "notifications": null,
54    "withheld_in_countries": []
55  }
```

Data distribution

5.3. EVALUATION

Table 5.4: Follower ranges and the number of tweets posted by users within the given range of followers.

Number of Followers	Tweets	%
0 - 49	10,358,082	13.78%
50 - 4,999	63,006,739	83.81%
5,000 - 9,999	806,565	1.07%
10,000 - 99,999	905,533	1.20%
100,000 - 999,999	90,620	0.12%
1,000,000+	9,457	0.01%

vers. This means that for the majority of tweets, this feature

User verified status

- Politicians, public figures, celebrities, journalists ...
 - Twitter to verify their profile
 - **'verified': True,**



```
13 "user": {
14   "id": 1161317586,
15   "id_str": "1161317586",
16   "name": "Nathan Fry",
17   "screen_name": "NathanFryLeeds",
18   "location": null,
19   "url": null,
20   "description": null,
21   "translator_type": "none",
22   "protected": false,
23   "verified": false,
24   "verified_type": "blue",
25   "followers_count": 119,
26   "friends_count": 299,
27   "listed_count": 1,
28   "favourites_count": 49823,
29   "statuses_count": 13594,
30   "created_at": "Fri Feb 08 22:08:57 +0000 2013",
31   "utc_offset": null,
32   "time_zone": null,
33   "geo_enabled": false,
34   "lang": null,
35   "contributors_enabled": false,
36   "is_translator": false,
37   "profile_background_color": "000000",
38   "profile_background_image_url": "http://abs.twimg.com/images/themes/theme1/bg.png",
39   "profile_background_image_url_https": "https://abs.twimg.com/images/themes/theme1/bg.png",
40   "profile_background_tile": false,
41   "profile_link_color": "1DA1F2",
42   "profile_sidebar_border_color": "000000",
43   "profile_sidebar_fill_color": "DDEEFF",
44   "profile_text_color": "333333",
45   "profile_use_background_image": true,
46   "profile_image_url": "http://pbs.twimg.com/profile_images/1544611925488705539/jukIhz0_normal.jpg",
47   "profile_image_url_https": "https://pbs.twimg.com/profile_images/1544611925488705539/jukIhz0_normal.jpg",
48   "profile_banner_url": "https://pbs.twimg.com/profile_banners/1161317586/1648019874",
49   "default_profile": true,
50   "default_profile_image": false,
51   "following": null,
52   "follow_request_sent": null,
53   "notifications": null,
54   "withheld_in_countries": []
55 }
```

User verified status : verifiedWeight

- Politicians, public figures, celebrities, journalists ...
 - Twitter to verify their profile
 - **'verified': True,**
 - Verification is a sign of authenticity
 - Roughly 290,000 accounts
 - 41% news, politicians, public figures, journalists

if (User['verified']):

 weight = 1.5

else:

 weight = 1.0

verifiedWeight= weight/1.5



Trivia - <https://help.twitter.com/en/managing-your-account/about-twitter-verified-accounts>

- Verification cost

**** Important Note:**

Twitter does not sell the blue Verification badge. A Twitter employee will never request financial compensation in exchange for a badge or as part of the application process. Twitter does not authorize any external agents or individuals to sell Verification on the platform. We encourage people to directly apply through the Verification application process available in the Twitter app. Individuals offering to buy or sell Verification are subject to loss of Verified status and permanent suspension from the platform.

How many verified users are on Twitter?

360,000 people

The new process creates a more public system for people to get verified on Twitter, which is an important form of social currency for pop culture stars, journalists, and world leaders alike. Currently, **more than 360,000 people** are verified on Twitter. 20 May 2021

Twitter Verification Has a Pricetag: \$15,000

Those little blue checks don't come cheap if you aren't a household name.

By Alexander Abad-Santos

The screenshot shows the Twitter Blue subscription interface. At the top, it says 'Blue' with a checkmark icon. Below that, a banner states: 'Blue subscribers with a verified phone number will get a blue checkmark once approved.' The main content area lists several benefits:

- COMING SOON**: Rocket to the top of replies, mentions and search. Tweets from verified users will be prioritized — helping to fight scams and spam.
- COMING SOON**: See half the ads. See 50% fewer ads than non-subscribers.
- NEW**: Post longer videos. You'll finally be able to post longer videos to Twitter.
- Get early access**: Get early access to select new features with Twitter Blue Labs.

At the bottom, there are two pricing options:

Annual Plan	Monthly Plan
SAVE 12% \$7.00 / month £84.00 per year billed annually	\$8.00 / month £96.00 per year billed monthly

A 'Subscribe' button is located at the bottom. Fine print at the very bottom states: 'By clicking Subscribe, you agree to our [Purchase Terms of Service](#). Subscriptions auto-renew until canceled, as described in the Terms. You can only subscribe on iOS and the Twitter website at this time. A verified phone number is required to subscribe. If you've subscribed on another platform, manage your subscription through that platform.'

Has default profile Image : profileWeight

```
"profile_background_tile": false,  
"profile_link_color": "1DA1F2",  
"profile_sidebar_border_color": "C0DEED",  
"profile_sidebar_fill_color": "DDEEF6",  
"profile_text_color": "333333",  
"profile_use_background_image": true,  
"profile_image_url": "http://pbs.twimg.com/profile_images/1522693473937285121/0",  
"profile_image_url_https": "https://pbs.twimg.com/profile_images/15226934739372",  
"profile_banner_url": "https://pbs.twimg.com/profile_banners/278177631/16014684",  
"default_profile": true,  
"default_profile_image": false,  
"following": null
```

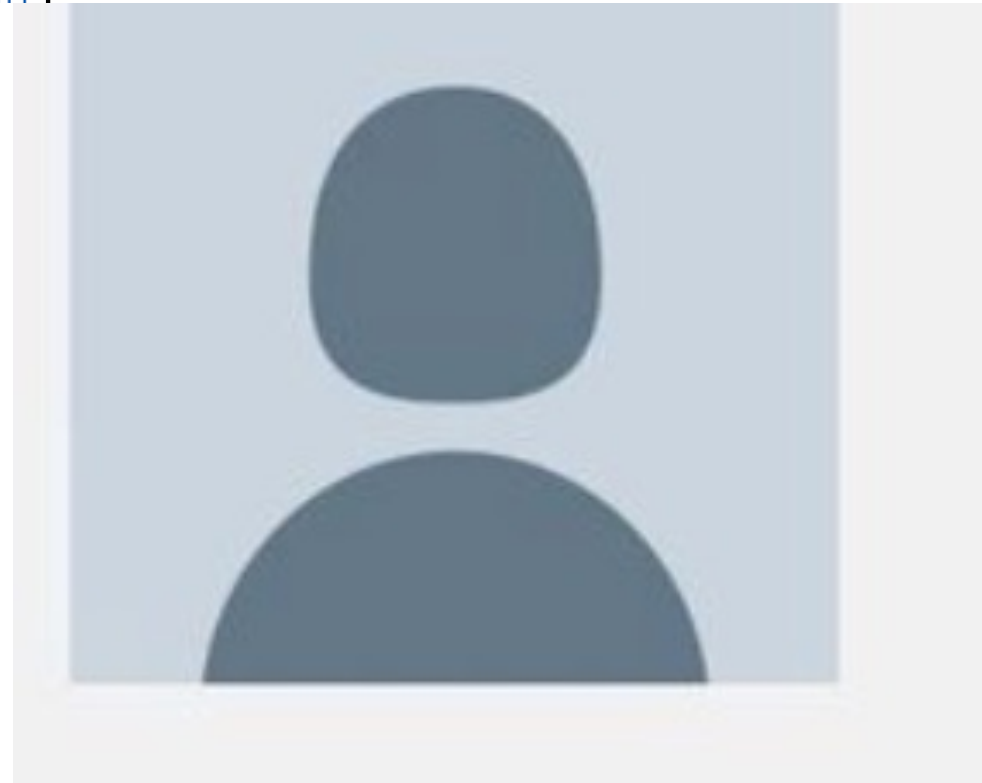
- 'default_profile': True,
- Users with default profile images are 'throwaway' accounts
 - Often created for the purpose of spamming or to post abuse

weight =1

if(User['default_profile_image']):

weight =0.5

profileWeight = weight



Quality Score : qualityScore

- qualityScore =

$$\frac{(\text{profileWeight} + \text{verifiedWeight} + \text{followersWeight} + \text{accountAgeWeight} + \text{descriptionWeight})}{5}$$

Range is [0 to 1]

- How to select high quality scores - thresholding
 - If qualityScore > 0.65 ? 
 - High quality text
 - If qualityScore < 0.45? 
 - Low-quality text

High Quality vs Low Quality Tweets

- Collect tweets with the following condition
 - If `qualityScore > 0.65` ?
 - High quality text
 - If `qualityScore < 0.45` ?
 - Low-quality text
- We get a set of potentially
 - High quality tweet texts
 - Low quality tweet texts
- A background model
 - Randomly selected from the stream

So far

- Newsworthiness
- We discussed creating a collection of, potentially!
 - High-quality and
 - Low-quality documents
- We applied Distant supervision
 - Using a set of heuristics created document set
 - High-quality set
 - Low-quality set
 - A random background set
 - Manually created thresholds
- Real-time
- adaptive

Newsworthiness Scoring model

Part 3

Newsworthiness scoring

- Documents are arriving sequentially
 - Background document model = a set of randomly selected tweets
- High quality Model
 - all documents in high quality text
- Low quality Model
 - All documents in low quality text
- Objective
 - Given a term, that it could appear in a certain group?
- **Let us define**
 - $R(t)$ = relative importance of term in a specific quality model
 - ***likelihood ratio for each term***
 - when compared to random background model
 - >1
 - Term is more common in the model than random
 - <1
 - Term is less common in the model than random

How could we create such weighting scheme?

How to compute a Likelihood ratio?

$$\bullet R_{HQ}(t) = \frac{P(\frac{t}{HQ})}{P(\frac{t}{BG})} = \frac{\frac{1+tf_{t,HQ}}{1+F_{BG}}}{\frac{1+tf_{t,BG}}{1+F_{BG}}}$$

- What values possible?

$$\bullet \frac{P(\frac{t}{HQ})}{P(\frac{t}{BG})}$$

- When it is > 1?
- When it is < 1?

- tf_t is term frequency of term t
- in respective group/class
- F is the raw frequency of all terms in the model
- $F = \sum f_t$

$$\bullet P(\frac{t}{HQ}) = tf_{t,HQ}/F$$

$$\bullet P(\frac{t}{BG}) = tf_{t,BG}/F$$

$$\bullet \frac{P(\frac{t}{HQ})}{P(\frac{t}{BG})}$$

Likelihood ratio

$$\bullet R_{HQ}(t) = \frac{P(\frac{t}{HQ})}{P(\frac{t}{BG})} = \frac{1 + tf_{t,HQ}}{1 + F_{HQ}} / \frac{1 + tf_{t,BG}}{1 + F_{BG}}$$

$$\bullet R_{LQ}(t) = \frac{P(\frac{t}{LQ})}{P(\frac{t}{BG})} = \frac{1 + tf_{t,LQ}}{1 + F_{LQ}} / \frac{1 + tf_{t,BG}}{1 + F_{BG}}$$

- HQ = High Quality model
- LQ = Low Quality Model
- BG = Background models

- tf_t is term frequency of term **t**
- in respective model
- **F** is the raw frequency of all terms in the model
- $F = \sum f_t$

>1

Term is more common in the model than random

<1

Term is less common in the model than random

```
if(rhQt > 2):  
    shQt = rhQt  
else:  
    shQt = 0
```

```
if (rlQt > 2):  
    slQt = rlQt  
else:  
    slQt = 0
```

cntd

$$\bullet S_{HQ}(t) = \begin{cases} R_{HQ}(t). & \text{if } R_{HQ}(t) \geq 2.0 \\ 0, & \text{otherwise} \end{cases}$$



$$\bullet S_{LQ}(t) = \begin{cases} R_{LQ}(t). & \text{if } R_{LQ}(t) \geq 2.0 \\ 0, & \text{otherwise} \end{cases}$$



- The idea is to prevent terms which have no clear association with either high quality content or low- quality content from affecting the overall content

Eco-warrior Greta Thunberg carried away by police at anti-coal mine demo <https://t.co/AjnMALu1F2>

• ['anticoal', 'away', 'carried', 'demo', 'ecowarrior', 'greta', 'mine', 'police', 'thunberg']

• **hqDFreq : 200665 lqDFreq: 227001 bgDFreq : 18442**

• anticoal

• tftbg : 0 tfthq : 0 tftlq : 0
• rhQt : 0.919084045548551 lhQt : 0.8124545706847107

• shQt : 0 lhQt : 0

• away

• tftbg : 137 tfthq : 164 tftlq : 186 8
• rhQt : 1.0989048370689198 lhQt : 1.1009348167973978
• shQt : 0 lhQt : 0

• carried

• tftbg : 2 tfthq : 10 tftlq : 2
• rhQt : 3.369974833678021 lhQt : 0.8124545706847107
• shQt : 3.369974833678021 lhQt : 0

• demo

• tftbg : 1 tfthq : 0 tftlq : 0
• rhQt : 0.4595420227742755 lhQt : 0.40622728534235536

• shQt : 0 lhQt : 0

```
if(rhQt > 2):  
    shQt = rhQt  
else:  
    shQt = 0  
  
if (rlQt > 2):  
    slQt = rlQt  
else:  
    slQt = 0
```

- ecowarrior
 - tftbg : 0 tfthq : 0 tftlq : 0
 - rhQt : 0.919084045548551 lhQt : 0.8124545706847107
 - shQt : 0 lhQt : 0
- greta
 - tftbg : 1 tfthq : 1 tftlq : 1
 - rhQt : 0.919084045548551 lhQt : 0.8124545706847107
- shQt : 0 lhQt : 0
- mine
 - tftbg : 33 tfthq : 26 tftlq : 49
 - rhQt : 0.72986085970032 lhQt : 1.1947861333598686
- shQt : 0 lhQt : 0
- police
 - tftbg : 88 tfthq : 121 tftlq : 44
 - rhQt : 1.2598680174935193 lhQt : 0.4107916368630559
 - shQt : 0 lhQt : 0
- thunberg
 - tftbg : 0 tfthq : 1 tftlq : 2
 - rhQt : 1.838168091097102 lhQt : 2.437363712054132
- shQt : 0 lhQt : 2.437363712054132

```

if(rhQt > 2):
    shQt = rhQt
else:
    shQt = 0

if (rlQt > 2):
    slQt = rlQt
else:
    slQt = 0

```



Newsworthiness score

$$N_d = \log_2 \left(\frac{1 + \sum_{t \in d} S_{HQ}(t)}{D} \mid \frac{1 + \sum_{t \in d} S_{LQ}(t)}{D} \right)$$
$$= \log_2 \frac{1 + \sum_{t \in d} S_{HQ}(t)}{1 + \sum_{t \in d} S_{LQ}(t)}$$

t refers to a term in document d

D number of terms

documents are newsworthy if we have a positive N_d

That is greater than 0

For the prior example

```
sHt += shQt  
sLt += slQt  
  
nScore = math.log2((1+sHt)/(1+sLt))
```

- sHt : 3.369974833678021
- sLt : 2.437363712054132
- nScore : 0.3463224581090705

Examples!

-
- @50sBornWomen @keirstarmer That might be problematic...
- ['@50sBornWomen', '@keirstarmer', 'might', 'problematic']
- @50sBornWomen tftbg : 0 tfthq : 0 tftlq : 0 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 0.8124545706847107
- shQt : 0 lhQt : 0
- @keirstarmer tftbg : 0 tfthq : 0 tftlq : 0 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 0.8124545706847107
- shQt : 0 lhQt : 0
- might tftbg : 121 tfthq : 112 tftlq : 45 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.8512827634998875 lhQt : 0.3063353299303007
- shQt : 0 lhQt : 0
- problematic tftbg : 3 tfthq : 3 tftlq : 0 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 0.20311364267117768
- shQt : 0 lhQt : 0
- sHt : 0 sLt : 0 nScore : 0.0

another

- Real Madrid vs Barcelona : Supercopa de Espania Final EL Clasico 2023 Live Link: Link 1: <https://t.co/0v6zeGhr9U> Link 2: <https://t.co/0v6zeGhr9U> Link 1: <https://t.co/JCHQfNycIB> Link 1: <https://t.co/D2B7g7dbSm> Link 1: <https://t.co/pHYMzugyZ4> sadddsadssd
- ['1httpstcod2b7g7dbsm', '1httpstcojchqfnycib', '1httpstcophymzugyz4', '2023', 'barcelona', 'clasico', 'espania', 'final', 'link', 'live', 'madrid', 'real', 'sadddsadssd', 'supercopa']
- 1httpstcod2b7g7dbsm tftbg : 0 tfthq : 0 tftlq : 0 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 0.8124545706847107
- shQt : 0 lhQt : 0
- 1httpstcojchqfnycib tftbg : 0 tfthq : 0 tftlq : 0 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 0.8124545706847107
- shQt : 0 lhQt : 0
- 1httpstcophymzugyz4 tftbg : 0 tfthq : 0 tftlq : 0 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 0.8124545706847107
- shQt : 0 lhQt : 0
- 2023 tftbg : 382 tfthq : 520 tftlq : 187 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 1.250242265615653 lhQt : 0.39880276576690754
- shQt : 0 lhQt : 0
- barcelona tftbg : 4 tfthq : 4 tftlq : 120 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 19.66140061057
- shQt : 0 lhQt : 19.66140061057
- clasico tftbg : 0 tfthq : 0 tftlq : 0 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 0.8124545706847107
- shQt : 0 lhQt : 0

- espania tftbg : 0 tfthq : 0 tftlq : 0 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 0.8124545706847107
- shQt : 0 lhQt : 0
- final tftbg : 34 tfthq : 69 tftlq : 22 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 1.838168091097102 lhQt : 0.5338987178785242
- shQt : 0 lhQt : 0
- link tftbg : 199 tfthq : 172 tftlq : 1023 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.7950076993994967 lhQt : 4.159767401905719
- shQt : 0 lhQt : 4.159767401905719
- live tftbg : 254 tfthq : 295 tftlq : 1321 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 1.066858343068122 lhQt : 4.21201938213799
- shQt : 0 lhQt : 4.21201938213799
- madrid tftbg : 8 tfthq : 14 tftlq : 302 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 1.5318067425809185 lhQt : 27.352637213051924
- shQt : 0 lhQt : 27.352637213051924
- real tftbg : 141 tfthq : 117 tftlq : 280 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.7637458970051341 lhQt : 1.6077446081859414
- shQt : 0 lhQt : 0
- sadddsadssd tftbg : 0 tfthq : 0 tftlq : 0 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 0.8124545706847107
- shQt : 0 lhQt : 0
- supercopa tftbg : 0 tfthq : 0 tftlq : 0 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.919084045548551 lhQt : 0.8124545706847107
- shQt : 0 lhQt : 0
- sHt : 0 sLt : 55.385824607665626 nScore : -5.817260609662692

another

- The minority speaker supporting anti choice and anti labor? Embarrassing
- ['anti', 'choice', 'embarrassing', 'labor', 'minority', 'speaker', 'supporting']
- anti tftbg : 99 tfthq : 93 tftlq : 53 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.863939002815638 lhQt : 0.4387254681697437
- shQt : 0 lhQt : 0
- choice tftbg : 25 tfthq : 41 tftlq : 24 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 1.4846742274245823 lhQt : 0.7812063179660679
- shQt : 0 lhQt : 0
- embarrassing tftbg : 10 tfthq : 3 tftlq : 15 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.33421238019947314 lhQt : 1.1817521028141247
- shQt : 0 lhQt : 0
- labor tftbg : 3 tfthq : 4 tftlq : 2 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 1.1488550569356888 lhQt : 0.609340928013533
- shQt : 0 lhQt : 0
- minority tftbg : 11 tfthq : 12 tftlq : 17 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.995674382677597 lhQt : 1.218681856027066
- shQt : 0 lhQt : 0
- speaker tftbg : 57 tfthq : 47 tftlq : 38 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.760621279074663 lhQt : 0.54630565959834
- shQt : 0 lhQt : 0
- supporting tftbg : 53 tfthq : 32 tftlq : 29 self.hqDFreq : 200665 self.lqDFreq): 227001 self.bgDFreq : 184428
- rhQt : 0.5616624722796701 lhQt : 0.45136365038039483
- shQt : 0 lhQt : 0
- sHt : 0 sLt : 0 nScore : 0.0